

Coupling-gated cortical coherence: what a field theory of conscious unity must assume, and what it predicts

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Abstract

A field account of cortical coordination holds that spatially distributed neural processes are coupled in part through the endogenous extracellular electromagnetic field. We ask what such an account must assume to reach a globally consistent, self-referential state, and what it predicts before it gets there. Its physical core is a noisy mean-field phase model whose coupling drifts through a synchronization threshold. If recovery of coherence after sleep is gated by that crossing, it is not free to take an arbitrary time course. For as long as the phase distribution tracks the stationary branch, it must show a square-root onset and a von Mises concentration–coherence relation with no fitted shape parameter. The tracking condition is part of the prediction, and the simulations measure its cost: ramp speed sets the apparent onset exponent, so an exponent fitted to a driven recovery trace reports the drive as much as the mechanism. A coupling plasticity stepping down a squared-drift objective removes most of the objective available to it at the instant each update acts, and keeps

the system coherent, while driving within-cluster coupling far below between-cluster coupling—away from the structure it is asked to learn, and further from it than from almost every relabelling of the nodes; two controls, each stepping at the size of its own arm’s gradient and the second carrying that gradient’s own entries to relabelled edges at an overlapping kernel deformation, do neither, and the same kernels are unremarkable against partitions of the same nodes drawn blind to frequency. Averaged over the intervals between updates the objective returns to its frozen level, and a bounded search over learning rates and update cadences finds coherence and lost alignment without a sustained reduction. Descending that functional does not produce representational learning: for this pairing of objective and template the optimum is opposed to the target by a conservation law rather than by accident. A Lean 4 formalization composes finite capacity, Landauer’s bound, the noisy Kuramoto transition, sheaf gluing and Banach’s theorem into one conditional theorem, isolating the eight hypotheses it consumes and the two structures it takes alongside them, whose class fields carry physical content of their own; one witness assembled from distinct toy components satisfies all eight at once. We defend as a philosophical commitment, not a theorem, the identification of the glued section with the unity of a conscious episode and of its reflexive fixed point with a minimal self. Testing the account requires independent calibration of effective cortical coupling and an awakening experiment separating slow coupling change from alternative recovery mechanisms—which a sequence of equilibrated phase distributions cannot do, since a coupling increase, a diffusion decrease and a matched common drive share every stationary density.

1 Introduction

Neural activity changes across sleep, awakening and anesthesia, but a change in synchrony does not by itself identify the physical interaction responsible for it. A field account makes a stronger proposal: spatially distributed neural processes are coordinated partly through the endogenous extracellular elec-

6 tromagnetic field, and changes in the medium can change that coordination
7 (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible
8 interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch,
9 2015). Whether that mechanism is sufficient to support the collective state
10 such accounts require remains an empirical question.

11 Such an account has three requirements, and they order what follows.
12 Coupling must be strong enough to sustain coherent collective activity; the
13 local descriptions that activity carries must be compatible enough to glue
14 into a single one; and that glued state must support an accurate model of
15 itself. The first requirement is physical and testable with current methods,
16 and most of this article is about testing it. The second and third are where
17 the mathematics is, and where the assumptions accumulate.

18 The first requirement admits a definite claim. If cortical coupling is slowly
19 gated—if recovery of coherence after sleep is driven by a coupling parameter
20 drifting through a synchronization threshold—then the recovery is not free
21 to take an arbitrary time course. For a recovery slow enough that the phase
22 distribution tracks the stationary branch, the theory fixes a square-root onset
23 and a one-parameter relation between phase concentration and coherence,
24 with no fitted shape parameter. That is a constraint an experiment can
25 meet or violate. How slow is slow enough is itself part of the claim, and
26 our simulations bound it from the other side, by showing which parts of the
27 constraint survive finite populations and finite ramp speeds and which do
28 not.

29 The contribution has four parts. The first is the coupling-gated recovery
 30 hypothesis itself: a stationary coherent branch with square-root onset, a von
 31 Mises concentration–coherence relation, and an experimental protocol with
 32 failure conditions declared in advance (Sections 2 and 3). The second is a
 33 constraint on how such data may be analyzed. The apparent onset exponent
 34 measured from a finite ramp is a function of ramp speed across the sweep
 35 of Section 4, so an exponent fitted to a recovery trace measures the ramp
 36 at least as much as the mechanism—and that holds for any analysis reading
 37 a critical exponent off a driven transition, not only for ours. The third is a
 38 negative result about plasticity: a coupling update stepping down a squared-
 39 drift objective removes most of the objective available to it at the instant
 40 it acts, and preserves coherence, while moving *away* from the environmen-
 41 tal structure it is expected to learn—below almost every node relabelling,
 42 below all but a twentieth of the partitions drawn blind to that structure,
 43 and beyond what either of two controls does: a random update scaled to its
 44 own arm’s gradient, and a gradient step delivered to relabelled nodes, whose
 45 cumulative kernel deformation brackets the gradient arm’s. The reduction
 46 is a property of that instant rather than of the run: averaged over the in-
 47 tervals between updates the objective sits at the frozen arm’s level, and a
 48 bounded grid of learning rates and update cadences, with criteria fixed in ad-
 49 vance, contains no case combining a sustained reduction with the coherence
 50 and the loss of alignment. Descending a squared-drift functional does not
 51 produce representational learning, and the direction of that failure follows

52 from a conservation law rather than from the particular run. The fourth
53 is a machine-checked composition (Section 5) in which finite information
54 capacity, Landauer’s bound, a predictive dissipation bound, the noisy Ku-
55 ramoto transition, sheaf gluing and Banach’s fixed-point theorem yield one
56 conditional theorem, with the eight hypotheses that composition consumes
57 isolated and named; one witness assembled from distinct toy components
58 satisfies all eight at once, so their conjunction is not empty.

59 A prose table of assumptions would list the same eight items, so the fourth
60 contribution needs a word about method. What the Lean development adds
61 is that it makes the list non-negotiable. The characteristic failure in an
62 argument of this shape is not a false step but a silent change of subject:
63 a theorem about a stationary density read as a claim about a trajectory,
64 an inequality about a register read as a claim about learning, agreement of
65 phases read as compatibility of representations. Type-checking forbids each
66 of those substitutions, and an assumption needed to make a step go through
67 appears as a hypothesis in the term rather than as a word in a sentence. The
68 last of them is not merely forbidden but refuted by a constructed object: the
69 development carries a cover at a single common phase whose local densities
70 disagree where its patches meet, satisfying every other requirement of the
71 gluing theorem and failing compatibility alone (Section 5.2). That object
72 is what a prose table of assumptions cannot carry. Formalization does not
73 make the assumptions true; it makes them impossible to lose.

74 The physical model and the formal composition belong in one article be-

75 cause they are two halves of one commitment. The coupling-gated prediction
76 is worth testing under this framework, rather than as one more synchroniza-
77 tion model, only because the coherent state it produces is claimed to do
78 further work: to support a globally consistent representation and a reflexive
79 fixed point that we identify with phenomenal unity and a minimal self.

80 The distinction that does that work is between phases agreeing and de-
81 scriptions fitting together, and it is concrete. Take two cortical regions whose
82 receptive territories overlap, each carrying a probability distribution over
83 what occupies the shared part. Their oscillators can be locked to the same
84 phase at every instant while one region's distribution on the overlap places an
85 object left of fixation and the other places it right. The phases agree; the de-
86 scriptions do not, and no single distribution over the shared territory restricts
87 to both. A theory that identifies unity with synchrony cannot mark that dif-
88 ference, and so cannot separate a brain whose regions oscillate together from
89 one whose regions are about the same world. The sheaf condition marks
90 exactly it: local states glue to one global section when their restrictions to
91 shared regions coincide, and otherwise not at all. Coherence enters as the
92 physical condition under which compatibility can be maintained, not as a
93 substitute for it, and the self-prediction map then asks the further ques-
94 tion of whether the glued state contains an accurate model of itself. That
95 identification is a commitment, not a theorem, and Section 6 states it, ar-
96 gues for it, and says what would count against it. Its separation from the
97 physical tests is what makes both assessable: the cortical coupling hypothe-

98 sis and the recovery prediction can fail while the mathematical composition
 99 stands, and the composition can stand while the identification is wrong. The
 100 established ingredients (Landauer, 1961; Still et al., 2012; Sakaguchi, 1988;
 101 Banach, 1922) therefore appear here for their biological composition, not for
 102 any new fundamental physical law.

103 **2 A coupling-gated model of cortical coher-** 104 **ence**

105 **2.1 Dynamics and stationary solutions**

106 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

107 Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and
 108 K_{ij} has inverse-time units, with angles treated as dimensionless. The uni-
 109 form mean-field convention is $K_{ij} = K/N$. Frequency heterogeneity and
 110 phase diffusion are separate parameters; a frequency-spread width cannot be
 111 substituted for D when applying the identical-frequency stationary theory.

112 For identical frequencies in their rotating frame, the assumed stationary

113 density is

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}. \quad (2)$$

114 The I_n are modified Bessel functions, and ψ is the collective phase. Lean
 115 verifies normalization, the density's circular order parameter, and the self-
 116 consistency equation. At $K \leq K_c = 2D$, the only nonnegative self-consistent
 117 order parameter is zero. At $K > K_c$, exactly one positive solution coex-
 118 ists with zero; the positive branch is continuous at threshold and strictly
 119 increasing with K .

120 The near-threshold expansion gives

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

121 This is a result about the stationary solution branch. Of the coherent branch
 122 we prove existence, uniqueness, continuous emergence at threshold and strict
 123 growth in K , but not that it is dynamically selected. The dynamical mean-
 124 field limit—propagation of chaos—is the missing link between the finite sys-
 125 tem and the stationary description (Sznitman, 1991), and no theorem in this
 126 development closes it. Exchange symmetry alone does not justify factoriza-
 127 tion of the joint law, particularly for an ensemble with an unpinned collective
 128 phase.

129 A spatial version replaces the finite sum by an integral against a kernel
 130 $K(x, y)$. We normalize the substrate measure to total mass one, so the mean-

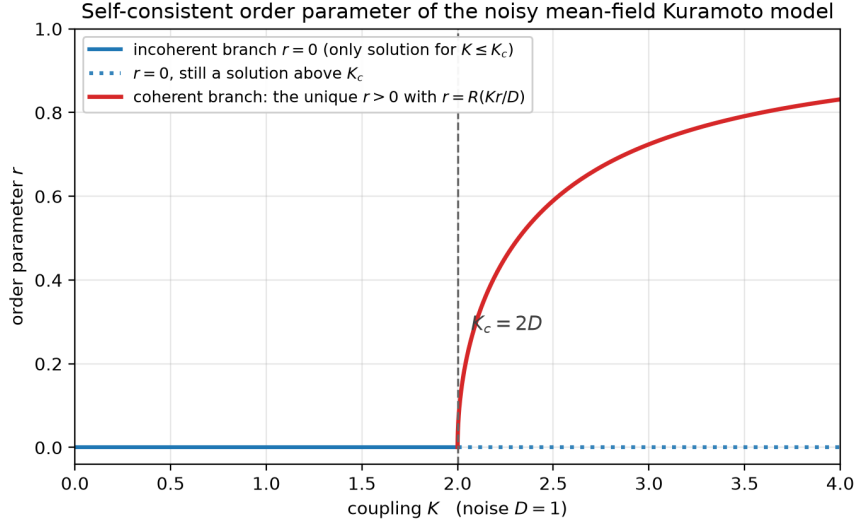


Figure 1: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the same integral expressions used in the formalization; it illustrates the static theorems and does not establish dynamical selection.

field strength is a kernel average, not a quantity that grows with the arbitrary
measure of the substrate. Reduction of a heterogeneous spatial kernel to a
scalar K is an additional modelling step, and the threshold in Eq. (2) does
not automatically apply to every spatial or signed network.

2.2 Physical identification and calibration

Regular mesh refinement gives convergence of a discrete coupling-energy
quadrature to its continuum integral: scalar energy convergence, not conver-
gence of trajectories and not a construction of a physical kernel. The model

139 posits a continuum kernel and separately identifies its cortical realization
 140 with endogenous electromagnetic coupling. The formal predicate requires
 141 a jointly continuous kernel, a normalized substrate, a mean-field strength
 142 equal to its spatial average, and no single site providing more than half of
 143 that mean-field coupling. Toy witnesses satisfy these conditions; none is a
 144 model calibrated to cortex.

145 Extracellular fields can modulate spike timing (Fröhlich & McCormick,
 146 2010; Anastassiou & Koch, 2015), and field-based models have been proposed
 147 for integrating distributed neural information (Pinotsis & Miller, 2022; Mc-
 148 Fadden, 2020). These observations motivate an effective interaction without
 149 establishing its magnitude relative to phase diffusion, its spatial averaging
 150 rule, or its causal role in experience; changes in phase organization under
 151 propofol (Bastos et al., 2021) likewise do not measure a crossing of the model’s
 152 coupling threshold.

153 The physical-unit conversion is the sharpest open limitation of the frame-
 154 work. An estimate proportional to $N\Delta tf$, using population size, a spike-time
 155 shift and oscillation frequency, is dimensionless. It cannot be compared di-
 156 rectly with an inverse-time diffusion parameter. A conversion such as

$$K_{\text{eff}} = \gamma N s E f \tag{4}$$

157 requires a rate calibration γ , where s is the time shift per unit field and E
 158 the field amplitude. Supplemental Material sets out why γ is not recoverable

159 from the other factors: it states how a per-cycle phase advance accumulates
160 into a drift rate, and the choice that would let it cancel removes the very
161 ratio the threshold test is about. The saved field-strength calculations are
162 conditional on this calibration and on the chosen spatial aggregation; they do
163 not independently establish $K_{\text{eff}} > 2D$ in cortex. Determining the interaction
164 kernel and phase diffusion from the same physical system is necessary before
165 the threshold becomes a quantitative biological claim; Section 3.2 makes that
166 measurement the first item of the proposed experiment.

167 **3 Recovery after awakening: a conditional** 168 **prediction**

169 Sleep-associated changes in extracellular space and clearance motivate a pos-
170 sible slow control of coupling (Xie et al., 2013). That geometry is measurable
171 and its cortical values are known: the extracellular space occupies a volume
172 fraction $\alpha \approx 0.2$ of tissue at a diffusion tortuosity $\lambda \approx 1.6$ (Syková & Nichol-
173 son, 2008), and natural sleep raises α from roughly 0.14 to 0.23 (Xie et al.,
174 2013). It reaches Eq. (4) through the field amplitude E rather than through
175 a further mechanism: below 10 kHz cortical tissue conducts approximately
176 resistively at $0.3\text{--}0.6\text{ S m}^{-1}$, an order of magnitude set by the conductiv-
177 ity of saline diluted into that volume fraction (Barbour, 2017), so at fixed
178 transmembrane current density a larger α yields a smaller extracellular field.
179 We hypothesize that recovery after awakening includes a change in effective

180 coupling $K(t)$, mediated by extracellular geometry, on a slower timescale
 181 than phase dynamics. The conductivity route fixes the sign that hypothesis
 182 predicts—contraction of the extracellular space on awakening raises K —so
 183 the design of Section 3.2 can fail it in direction as well as in magnitude. It
 184 fixes nothing further: it is an effective-medium argument at fixed current
 185 density, and is silent on how a heterogeneous field aggregates into the scalar
 186 K of Eq. (2). Neither the monotonic relation between geometry and cou-
 187 pling nor its role in human sleep inertia follows from the cited physiology;
 188 the model requires both to be measured.

189 If $K(t)$ rises through K_c and the phase distribution tracks the coherent
 190 stationary branch, Eq. (3) predicts

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

191 The ideal stationary description has a subthreshold interval followed by con-
 192 tinuous onset. Saturation requires a specified subsequent coupling trajectory;
 193 monotonic coupling alone does not determine a logistic or any unique sig-
 194 moid. The onset time is fixed by $K(t_0) = K_c$ only if $K(t)$ is independently
 195 known. When coupling is inferred from the same coherence trace, this is not
 196 a parameter-free prediction of latency.

197 Critical slowing down limits the adiabatic prediction of Eq. (5) near
 198 threshold, and a finite population carries nonzero fluctuating order below
 199 it. An observable onset therefore requires a predeclared threshold above the

finite-size floor, and a finite-rate trajectory need not show an infinite-slope cusp. Section 4.1 quantifies the limitation.

3.1 Distributional and temporal tests

The stationary model makes a second prediction: phase concentration and coherence obey $r = I_1(a)/I_0(a)$, with no fitted shape parameter. Its applicability depends on the stationary-density assumption and the observation model, and the relation is common to the whole von Mises family, so agreement with it identifies neither ephaptic coupling nor this framework against alternative synchronization models. Within the same ansatz $a/r = K/D$ for $r > 0$, so the limiting threshold ratio follows from the density relation instead of testing it, and ratios near the finite-size floor are especially unstable.

The estimators must be independent, which is the constraint most easily violated in practice. Estimating a by inverting the Bessel ratio from r would enforce the relation being tested, and linear statistics on wrapped phases estimate neither quantity.

3.2 A protocol for the awakening experiment

The requirements above define a design rather than a single measurement, and the design is where the content of the hypothesis lives. An adequate test needs all six of the following.

1. **Coupling and diffusion in the same units.** Estimate K and D

from the same cortical regime, in inverse-time units, separating phase diffusion from quenched frequency spread and from observation noise. Without this the threshold $K > 2D$ is not an evaluable statement about cortex. A sequence of equilibrated phase distributions will not supply it: a coupling increase, a diffusion decrease and a common drive of matched strength give identical stationary densities window for window, so the estimate must reach the phase increments that carry the rate, and telling an endogenous coupling from a common drive needs a measured drive or a controlled perturbation on top of that (Supplemental Material).

2. **Geometry calibrated to coupling.** A diffusion-MRI observable or an astrocytic signal is an indicator of the extracellular volume fraction α , not a measurement of K . Its mapping to effective coupling must be established separately, with its own uncertainty, and the conductivity argument of Section 3 supplies only the expected sign of that mapping.

3. **Phase estimation at adequate spatial resolution.** At least 100 spatial samples per estimate. Estimate r by the circular resultant length and a by an unweighted regression of log-density on the cosine of phase relative to the circular mean. With fewer sites, pooling over time changes the estimand to a spatiotemporal distribution and must be stated as such.

4. **An independent recovery endpoint.** Behavioral recovery measured

242 separately from the phase trace, so that latency is not defined by the
243 signal whose latency is at issue.

244 **5. A predeclared onset criterion.** A coherence threshold fixed above
245 the finite-size floor before unblinding, chosen in the knowledge that a
246 finite ramp smears the cusp.

247 **6. Model comparison under matched observation.** Coupling-gated
248 recovery compared against exponential, flexible sigmoidal and other
249 slow-recovery models under identical temporal filtering and observation
250 noise. Latency alone cannot distinguish the mechanism, because many
251 processes create delays; neither can a coherence trace alone, which the
252 mechanisms of item 1 share exactly.

253 The design fails informatively in three ways: failure of the stationary
254 phase-density relation over a sufficiently broad concentration range; absence
255 of the proposed coupling–geometry association; or a recovery trace better
256 explained by an alternative after finite-rate and observation effects are ac-
257 counted for. Two apparent failures are not failures. A linear apparent onset
258 is evidence against the adiabatic approximation only where that approxima-
259 tion is independently justified, and it does not falsify the field hypothesis
260 under arbitrary ramp speeds. The absence of a fold in the stationary equa-
261 tion excludes stationary coexistence of two positive branches in this model,
262 but does not exclude finite-rate hysteresis or adjudicate theories of ignition.

263 4 Simulation results

264 The simulations characterize the finite-size, finite-rate regime that the sta-
265 tionary theory leaves open, and they bear on whether the prediction of Sec-
266 tion 3 is measurable at all. Two of them, Sections 4.1 and 4.3, illustrate
267 methodological issues that also arise in analyses of driven transitions and
268 dissipation-driven plasticity. Each is a finite run at its sampled parameters
269 and not a general statement, and none discharges a hypothesis of the com-
270 position in Section 5. Each run uses explicit seeds and compact saved sum-
271 maries, and the numerical values below are generated from those artifacts.
272 The supplement gives parameters, estimators, failure criteria and additional
273 panels. Identical-frequency runs use phase diffusion rather than a quenched
274 frequency-spread surrogate, and order plots display the $1/\sqrt{N}$ reference.

275 4.1 Ramp speed sets the apparent onset exponent

276 The dynamic-ramp study (Fig. 2) uses $N = 2000$, $D = 1$ and 32 replicas at
277 four coupling speeds. Escape is the first of three consecutive saved samples at
278 $r \geq 0.2$ after threshold. Only 5/32 replicas escape within the fastest window.
279 The three slower ramps have mean coupling delays 0.1947, 0.0684 and 0.0253,
280 with a fitted delay exponent 0.443. Their apparent onset exponents run from
281 0.898 to 0.417, against 0.443 to 0.448 returned by the same estimator on the
282 exact stationary branch sampled on the same coupling grids. At the fastest
283 speed the ensemble mean never reaches the declared fit window, so the value

the estimator returns there, 0.964, is a local log-slope of a delayed trajectory rather than an onset exponent.

The exponent recovered by fitting a driven transition is a joint function of the mechanism and the drive rate. An onset exponent read off a recovery trace therefore says little about the underlying transition unless the ramp rate is independently known and the finite-size floor is declared, and that holds for any such analysis, not only for one conducted under this framework. The prediction of Section 3 is the first thing it constrains: the stationary square-root law cannot be imposed on an arbitrary recovery trace, and its absence is not by itself evidence against a coupling-gated mechanism. The specified deviation-based critical ramp speed is not bracketed by the sweep.

The dynamical-selection control shows escape toward a coherent branch above threshold. At $K = 2.8$, steady mean order is 0.67735 against the static value 0.68270. Refining the integration step by a factor of 4 at fixed duration moves that order to 0.68168, so the bulk of the gap is Euler–Maruyama bias; the last refinement still moves it by 0.00147, so the remaining -0.00102 bounds the finite-size part rather than measuring it. The fitted growth-rate slope is 0.41505 against the theoretical $1/2$, over a window bounded below by the finite-size floor and above by half the static branch order rather than by the extent of the linear regime; that window, finite size and the integrator all contribute to the shortfall, and this sweep separates none of them. The propagation-of-chaos diagnostics show decreasing subcritical dependence with increasing population size. In the supercritical unpinned ensemble, how-

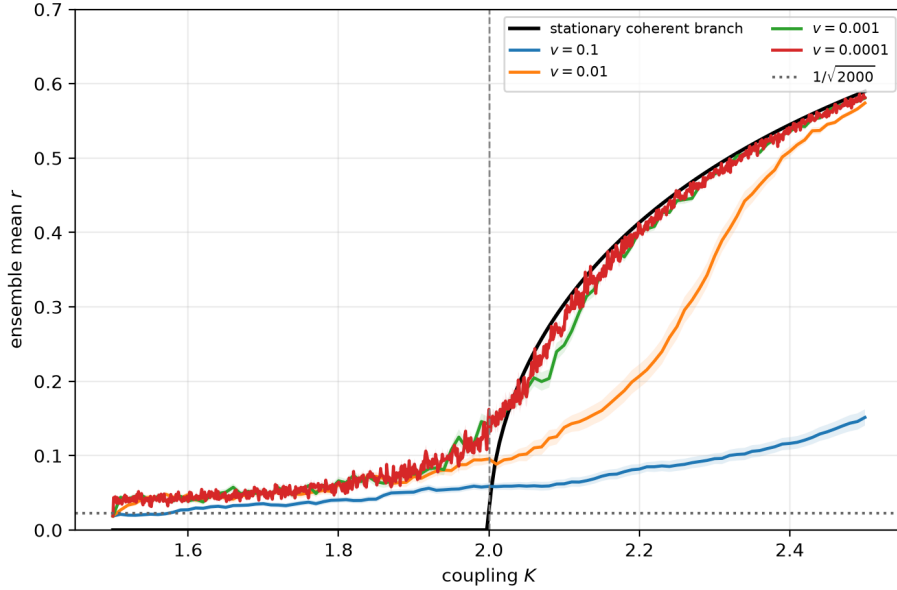


Figure 2: Finite-rate departures from the stationary branch. Ensemble-mean order is plotted against the increasing coupling for four ramp speeds v , with bands of one standard error across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape; the stationary branch is not a universal time course for recovery.

307 ever, pair correlation remains substantial because realizations share a random
 308 collective orientation. This negative unconditional result does not settle a
 309 conditional mean-field limit after removal of that orientation.

310 4.2 Spatial coupling, frustration and calibration

311 A periodic two-dimensional sheet with row-normalized exponential coupling
 312 remains coherent in the sampled empirical decay-length band under the de-
 313 clared total coupling and frequency spread. At decay lengths 0.1, 0.2 and

0.3 mm, steady order is 0.9424, 0.9427 and 0.9427. This is one finite parameter regime; fixing total coupling while changing range tests fragmentation, not whether cortex supplies that total coupling.

A row-balanced Dale-law network shows that row balance alone does not guarantee a floor-level incoherent baseline. In a separately specified weak-synapse regime, a uniform modulatory term produces an operational coherence crossing inside the bracket $K_{\text{eff}}/D \in [1.958, 2.009]$, from a coupling grid refined to steps of 0.025 inside the one geometric step $[1.857, 2.110]$ that locates it. The refined bracket still contains the model’s own mean-field threshold $K_c/D = 2$, so a tenfold finer grid does not separate the crossing from the threshold at this size. Signed coupling amplifies finite-size fluctuations in the same runs, whose incoherent baseline carries $N\langle r^2 \rangle > 1$, which is the direction a resolved crossing would be displaced in; the refinement bounds that displacement by the width of the bracket rather than exhibiting it. The quantity K_{eff}/D is dimensionless; carrying it over to cortex needs the rate calibration γ of Section 2.2, which neither this sweep nor the spatial control supplies.

4.3 Squared-drift plasticity does not learn structure

The predictive dissipation bound of Section 5 bounds nonpredictive memory by dissipation, and it is tempting to read a system that lowers its dissipation as thereby coming to mirror the structure of its input. The bound licenses no such reading: a memory carrying no information at all satisfies it with room

336 to spare, so lowering dissipation presses toward less representation rather
 337 than toward better-aligned representation. This control exhibits a system in
 338 which the pressure is not merely toward less but away.

339 For fixed phases, consider the squared-drift objective

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (6)$$

340 It is a quadratic functional of an affine image of the kernel. The formalization
 341 constructs the bounded drift operator A and its adjoint on the relevant L^2
 342 spaces, obtaining $\nabla\sigma = (2/D)A^*(AK + \omega)$, and the specified gradient flow
 343 decreases this objective at fixed phases. The objective is not generally a
 344 thermodynamic entropy-production rate, which requires probability-current
 345 information. Whether biological couplings follow this flow is an additional
 346 modelling question.

347 The joint control evolves noisy phases with clustered frequencies and
 348 nonzero grand-mean drive, while updating symmetric couplings every 50 fast
 349 steps. Couplings remain nonnegative, hollow and normalized to a fixed to-
 350 tal. The symmetric-edge derivative of $\frac{1}{2} \sum_i v_i^2$ is $(v_i - v_j) \sin(\theta_j - \theta_i)$, so both
 351 endpoints contribute. Resource constraints and evolving phases remove the
 352 fixed-phase monotonicity guarantee. Each seed runs four arms on one phase-
 353 noise stream: the gradient step, a symmetric random step at the Frobenius
 354 norm of that arm's own gradient, that arm's own gradient step under a ran-
 355 dom node relabelling, and a frozen kernel; every arm computes its direction

356 from its own phases and couplings, so the controls are set up like the gra-
 357 dient arm rather than fed its step. The reported quantity is how much of
 358 the objective the update removes at the instant it acts, since diagnostics are
 359 sampled at the update cadence, immediately after each update: symmetric
 360 coupling cancels from the mean drift, which fixes a floor of 990 beneath the
 361 frozen arm’s 1380.6063 to 1382.3448, and on that readout the gradient arm
 362 removes between 0.6773 and 0.6914 of that headroom. The random arm,
 363 stepping at the size of its own gradient, removes none of it, so the reduction
 364 belongs to the gradient direction. A step size set the same way does not
 365 accumulate the same deformation: successive isotropic steps cancel, leaving
 366 the random arm at a cumulative norm growth of 1.2879 to 1.3056 against the
 367 gradient arm’s 12.4094 to 14.6958, so it barely enters the channel through
 368 which clip-and-rescale concentrates the fixed resource onto fewer edges. The
 369 permuted arm enters that channel. A relabelling is an isometry that leaves
 370 the multiset of step entries untouched, so the arm takes a step of its own
 371 gradient’s size and shape on edges that gradient did not select: it reaches
 372 a norm growth of 11.4185 to 16.3296, bracketing the gradient arm’s, and a
 373 second-half order of 0.9184 to 0.9189, which lies inside the gradient arm’s
 374 range. Its descent fraction is -0.3319 to -0.1607 : over deformation and co-
 375 herence ranges that overlap the gradient arm’s, a step carrying a gradient’s
 376 own entries to the wrong edges raises the objective instead of lowering it.
 377 The learning rate 0.2 is the rate at which that readout falls furthest across a
 378 sweep from 0.001 to 0.4 evaluated on one of the three reported seeds, not a

379 rate at which the update is known to converge.

380 Across three seeds, second-half order lies between 0.9167 and 0.9222: co-
381 herence is maintained throughout. Alignment with the cluster template is
382 read from the ratio of mean within-cluster to mean between-cluster cou-
383 pling, which is unchanged by rescaling the kernel. That ratio starts between
384 0.9824 and 0.9948 and ends between 0.3507 and 0.6135, while the random
385 arm leaves it at 0.9965 to 0.9997 and the comparably deformed permuted
386 arm at 0.7290 to 1.2935, a range that contains the no-preference line and
387 does not overlap the gradient arm's; against 2000 node relabellings the final
388 kernel is further from the true template than from a fraction 0.9510 to 1.0000
389 of them (Fig. 3). The reference for that fraction is the frozen arm on the
390 same seeds, which reaches 0.6435 to 0.9110 without any update at all, and
391 not a null median of one half; the increment the gradient step contributes
392 is the difference between those ranges, which at three seeds is small enough
393 that they nearly meet. The permuted arm reaches 0.2680 to 0.8410, so what
394 carries the gradient arm above the frozen reference is the direction of the step
395 and not the deformation it accumulates. What the update moves against is
396 that partition and not block structure at large. Read against 2000 partitions
397 of the same group sizes drawn blind to frequency, at most a fraction 0.0495
398 of them give a lower within-over-between ratio than the frequency partition
399 does on any seed, where the frozen arm's untouched kernel gives 0.0755 to
400 0.3480; and one such partition, interleaving the nodes so that each group
401 draws equally from all three clusters, leaves the ratio at 0.8071 to 3.0223,

402 inside that family and on both sides of the no-preference line. The update
 403 is opposed to the structure the environment carries rather than indifferent
 404 to structure. Two readouts do not carry the effect and we do not rest it
 405 on them. The initial-minus-final Frobenius distance to the template lies be-
 406 tween -58.0962 and -48.1760 , so that distance grows; but the kernel norm
 407 grows by a factor of 12.4094 to 14.6958 over the same interval, and a fixed
 408 total resource constrains the sum of the entries rather than the norm, so
 409 most of that growth is concentration of mass onto fewer edges rather than
 410 loss of alignment. The edge-level correlation with the template ends between
 411 -0.0362 and -0.0208 , near its starting value: at this sparsity the edge-level
 412 variance swamps a contrast that the block means resolve. The template is a
 413 covariance of shared latent cluster signals, not an input to the update or a
 414 measured covariance of the constant frequencies.

415 Most of the available objective is removed at the sampled instants, coher-
 416 ence is maintained, and within-cluster coupling falls to a fraction of between-
 417 cluster coupling. Coherence is therefore not a proxy for representational
 418 alignment in this regime, and a plateau in the objective establishes neither
 419 stationarity of the adaptive process nor a thermodynamic entropy-production
 420 rate.

421 What the update takes off the objective at the instant it acts, the interval
 422 that follows returns. Measured at every integration step and averaged over
 423 complete update intervals, a run at this rate and cadence sits at 0.998 and
 424 1.003 of its frozen arm across the two halves of its second half. A bounded

425 study fixed in advance asks whether some other declared regime holds a re-
 426 duction while keeping coherence and losing alignment: 6 combinations of
 427 learning rate and update interval, each against a paired frozen arm, scored
 428 on complete intervals against criteria and a selection rule declared before
 429 the runs. None reaches the declared 5% sustained reduction, the best falling
 430 between 1.22% and 1.36%, while coherence and the loss of alignment ap-
 431 pear across the grid. This regime therefore exhibits a coupling update that
 432 loses environmental structure while remaining coherent, and not one that
 433 sustains a lower objective; the second is what an argument from dissipa-
 434 tion minimization to representation would need, and a bounded grid that
 435 does not supply it does not show that nothing does. Nor is the direction
 436 of the failure accidental. Symmetric coupling conserves the mean drift, so
 437 minimizing $\sum_i v_i^2$ can only equalize drift across nodes, and equalizing drift
 438 rewards coupling between clusters of unlike frequency—the complement of
 439 a template that rewards coupling within clusters of like frequency. The ob-
 440 jective and the template are in this pairing opposed by construction, and
 441 that is the point: nothing about a system’s dissipation fixes which struc-
 442 ture its slow variables acquire. What generalizes beyond the pairing is that
 443 last sentence and not the sign of the effect, which is why the blind-partition
 444 reading matters—the update locates the environment’s own structure and
 445 moves against it, rather than degrading any structure a reader might score
 446 it against. What fails here is the objective rather than thermodynamically
 447 specified learning at large: an Ising model whose couplings are annealed de-

448 grees of freedom, driven by external fields that carry the data, memorizes and
449 generalizes under ordinary statistical-mechanical rules (Corberi et al., 2026).
450 Those couplings equilibrate under a Hamiltonian containing the data; these
451 descend a squared-drift functional whose stationary point the data never en-
452 ters. Which functional the slow variables follow is what decides whether they
453 acquire environmental structure.

454 **4.4 The estimation protocol demonstrated on public** 455 **EEG**

456 Item 3 of Section 3.2 specifies estimators whose behavior on real recordings
457 is not obvious, so we exercise them on public propofol-sedation EEG (Bajwa
458 et al., 2025). The purpose is to show what the protocol produces and where
459 it is fragile, not to test the awakening hypothesis, for which these data are
460 the wrong design.

461 Two fragilities appear immediately. The implemented estimator pools 62
462 channels over a 100-ms window, so it estimates a spatiotemporal distribu-
463 tion rather than an instantaneous spatial one—a departure from the protocol
464 forced by channel count, and one that any scalp implementation will face.
465 Bipolar and common-reference montages give different residuals, making ref-
466 erencing a material analysis choice rather than a preprocessing detail.

467 The bipolar results are compatible with the Bessel relation over the ob-
468 served concentration range. That range is too narrow to distinguish the

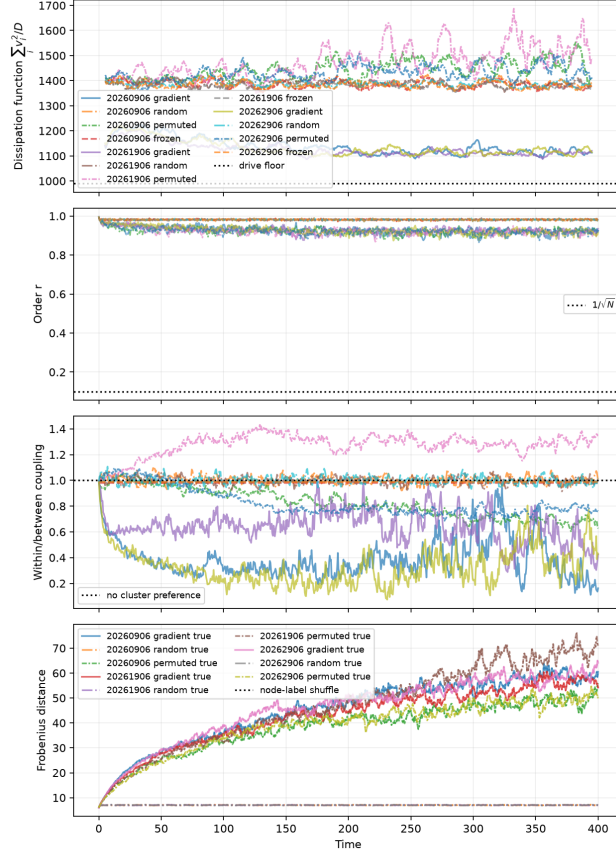


Figure 3: Joint fast/slow dynamics, four arms per seed on one phase-noise stream: gradient (solid), random step at its own gradient’s norm (dash-dotted), the gradient step under a random node relabelling (dash-dot-dotted), frozen kernel (dashed). First: smoothed squared-drift dissipation, sampled just after each coupling update, against the drive floor the conserved mean drift imposes; only the gradient arm falls, and only on that readout. Second: order and the finite-size reference. Third: mean within-cluster over mean between-cluster coupling, a measure unchanged by rescaling the kernel; the gradient arm falls well below the no-preference line, the relabelled arm stays above the gradient arm at an overlapping kernel deformation, and the remaining arms stay on the line. Fourth: Frobenius distance to the true template (solid) and to a node-label shuffle (dotted), which track each other because the kernel norm, not the alignment, is what grows. Coherence persists while alignment is lost. The result limits the proposed connection between dissipation minimization and environmental structure.

469 curve from its linear approximation at the observed scatter, so the check is
 470 uninformative about the relation itself. An informative test needs a broader
 471 concentration range and explicit treatment of source mixing, finite samples
 472 and temporal pooling; intracranial or high-density recordings across a gen-
 473 uine state transition are the natural setting.

474 5 Thermodynamic constraints and the for- 475 mal composition

476 The synchronization model of Section 2 is one link of a longer conditional
 477 construction, whose nine nodes are:

478 **n1** a finite phase space has bounded information capacity; **n2** a
 479 broken symmetry forces a boundary, where the field leaves its vac-
 480 uum manifold; **n3** the boundary erases, and erasure costs heat;
 481 **n4** that dissipation bounds nonpredictive memory; **n5** discrete
 482 couplings coarse-grain to a continuum limit; **n6** a specified phys-
 483 ical field realizes that limiting coupling; **n7** above threshold the
 484 field carries a unique coherent order parameter; **n8** compatible
 485 local states glue to a unique global section; **n9** that section is the
 486 unique fixed point of a self-prediction map.

487 Node-level results are established or cited; what the composition consumes
 488 are the eight implications between them. Figure 4 shows where they sit and
 489 what kind of assumption each one is, and Table 1 states them, classified by
 490 their role in the composition rather than by their plausibility. In particular,
 491 finite capacity does not imply a symmetry-breaking boundary: n1 and n2 are

492 independent roots, even though the composition can accept an implication
493 between them as an argument.

Edge	Status	Required connection
E12 capacity $\rightarrow$ boundary	Independent physical premise	A finite-capacity system realizes a field that leaves its vacuum manifold. Capacity alone does not imply this.
E23 boundary $\rightarrow$ erasure	Formalization gap	The boundary is associated with a finite register whose update leaves states unreachable.
E34 erasure $\rightarrow$ prediction	Formalization gap	Erasures supplies a predictive structure with the same thermal scale and heat budget, and nonzero wasted memory.
E45 prediction $\rightarrow$ continuum	Modelling assumption	A predictive dissipation bound places coupling energies in a convergent coarse-graining regime.
E56 continuum $\rightarrow$ field	Physical commitment	The limiting coupling is realized by a specified field satisfying the kernel conditions. Its identification with cortex is empirical.
E67 field $\rightarrow$ coherence	Physical commitment	That field operates above the synchronization threshold.
E78 coherence $\rightarrow$ unity	Formalization gap	Coherence drives a cover, supplied as a separate argument of the theorem, to an equilibrium reached by relaxation at coupling at least the named mean-field value. That its patches carry compatible local states is a field of the cover rather than a conjunct of this edge.
E89 unity $\rightarrow$ self	Modelling assumption	The state glued on that same cover is fixed by a contracting map mediated by its local self-encoding.

Table 1: The assumptions consumed by the conditional composition theorem, each an implication between two of the nine nodes above. Node-level results do not establish these connecting implications. The complete theorem map and witnesses are in Supplemental Material, Table S1.

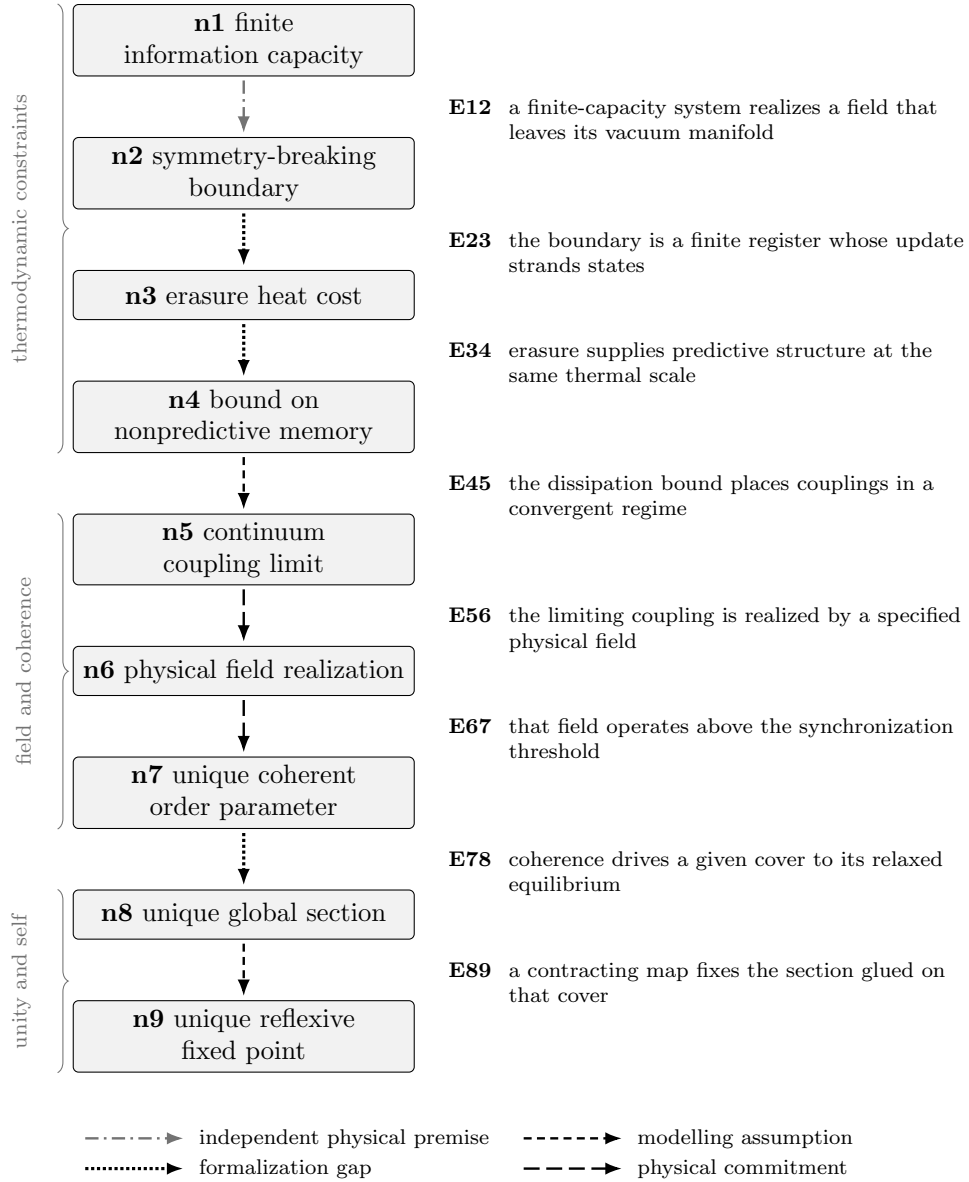


Figure 4: The conditional composition. Each node is an established or cited result; each arrow is an implication the composition consumes and no theorem here supplies, styled by the kind of assumption it carries. The three formalization gaps bracket the two physical commitments, so the chain’s empirical content sits between links of a different kind on either side. E12 is drawn as a premise because n1 and n2 are independent roots. Table 1 states each connection in full.

494 5.1 Capacity, boundaries and predictive memory

495 For a finite state space X , Shannon entropy satisfies $H(p) \leq \log |X|$, with
 496 equality only for the uniform distribution. A stream with more possible
 497 histories than $|X|$ cannot be encoded injectively in that register. Separately,
 498 a continuous field on a connected substrate that reaches distinct components
 499 of a disconnected vacuum manifold must leave that manifold somewhere,
 500 the simplest topological-defect construction (Kibble, 1976). Relating this
 501 locus to a particular finite information-processing register is the additional
 502 hypothesis E12 carries.

503 A non-surjective update on a finite register is non-injective. Under the
 504 specified system–bath model and thermal assumptions, erasure carries a Lan-
 505 dauer heat cost (Landauer, 1961). Finite capacity does not imply that every
 506 update erases, and logical many-to-one processing must be distinguished from
 507 its physical implementation (Norton, 2005; Shenker, 2000). The formaliza-
 508 tion records the physical heat identity as an obligation of each model and
 509 supplies an explicit one-bit eraser satisfying it.

510 For a system state X_t driven by an environmental signal S_t , the predictive
 511 dissipation bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (7)$$

512 Under the no-back-action Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$, data process-
 513 ing makes the bracket nonnegative (Still et al., 2012). The bound constrains

514 nonpredictive memory at a given dissipation budget. It does not specify
 515 a learning rule, maximize prediction, or require the internal representation
 516 to reproduce environmental geometry—the point the plasticity control of
 517 Section 4 makes concretely. Connecting this budget to Landauer heat ad-
 518 ditionally requires nonpredictive information to be bounded by the entropy
 519 erased by the register. The formal bridge derives Eq. (7) from Landauer’s
 520 bound under that identification; it does not derive the identification.

521 A bound $D_{KL} \leq \Delta t \sigma$ also cannot force $D_{KL} \rightarrow 0$ unless its right-hand
 522 side tends to zero, and Gibbs’ inequality proves nonnegativity of relative en-
 523 tropy, not a general identification of relative entropy with physical entropy
 524 production. The chain uses a different technical sense of information at each
 525 step—a counting bound on a register, a thermodynamic heat cost, a predic-
 526 tive mutual information—and those senses are not interchangeable (Rosas
 527 et al., 2026). These distinctions prevent an informational inequality from
 528 being read as a derivation of environmental learning, Bayesian inference, or
 529 semantic content.

530 **5.2 Compatible states and reflexive fixed points**

531 A sheaf formalizes local descriptions and their restriction to smaller regions.
 532 If local sections agree on every overlap, the sheaf condition gives a unique
 533 global section. In the probability-sheaf construction, this is consistency of
 534 local density information, and it is not implied by agreement of oscillator
 535 phases: the development includes a counterexample with equal phases but

536 incompatible densities on an overlap. Section 6 rests on this distinction,
537 and the model excludes such configurations only through its compatibility
538 hypothesis.

539 A thermodynamic cover couples this gluing construction to a phase con-
540 figuration at a potential minimum. For the deterministic positive-coupling
541 model, convergence to that minimum is proved under an explicit initial basin
542 condition; other equilibria prevent an unconditional conclusion. The station-
543 ary noisy order parameter, the deterministic relaxation, and the cover must
544 therefore be connected by E78 rather than identified without a hypothesis.
545 Neither field continuity nor synchronization alone guarantees a globally com-
546 patible representation.

547 A reflexive extension equips global sections with a complete metric and a
548 predictive map formed by restricting a state to a local encoding region and
549 reading that encoding out. If the map contracts and fixes the section supplied
550 by the cover, Banach’s theorem gives uniqueness of that same fixed point.
551 The contraction rate is assumed, and the equality between the glued section
552 and the fixed point is part of E89. A coherent order parameter supplies the
553 supercritical inequality required by the stated contraction bound; it does not
554 construct the map or prove that cortical dynamics implements it.

555 The theorem `chain` concludes `UnifiedSelf` from these hypotheses and
556 from two structures it also takes as arguments, a thermodynamic cover and
557 a reflexive boundary. The eight edges are not the whole of what is assumed:
558 the cover’s overlap-compatibility field—the load-bearing assumption of the

559 unity-is-not-synchrony argument—and the boundary’s resonance condition
 560 are class fields, so they are consumed by the theorem without appearing
 561 among its edges. A single witness satisfies all eight at once, so their conjunc-
 562 tion is not empty; it is assembled from a double well, a register, a uniform-grid
 563 mesh and a three-site cortex, which remain distinct toy systems rather than
 564 one mechanism. The project declares no additional Lean axioms beyond
 565 the foundational axioms used by Lean’s libraries; physical content resides
 566 in hypotheses and model fields. This checks the conditional mathematical
 567 composition, not the empirical truth of its premises. Detailed constructions,
 568 scope statements and counterexamples are retained in the supplement and
 569 source files.

570 **6 What the construction commits to about** 571 **consciousness**

572 The composition terminates in a mathematical object, and we take a posi-
 573 tion on what that object is. The commitment has two clauses. *Unity*: the
 574 unique global section glued from compatible local states is the formal corre-
 575 late of the unity of a conscious episode—the fact that a subject undergoes
 576 one experience rather than a collection of co-occurring local ones. *Selfhood*:
 577 the fixed point of the self-prediction map on that same section is the formal
 578 correlate of a minimal self, a state that is what it takes itself to be. Neither
 579 clause is a theorem, and no simulation here supports either. We argue for

580 them, then say what would count against them.

581 Three features of the construction motivate the identification, and each
582 distinguishes it from a bare synchrony measure.

583 First, compatibility rather than agreement. Oscillator phases can agree
584 everywhere while the local probabilistic descriptions they carry remain in-
585 compatible on their overlaps, and Section 5.2 exhibits that configuration
586 inside the formalization, not only as an illustration. If the unity of experi-
587 ence is a matter of local contents being about one world rather than merely
588 co-timed, then compatibility, not coherence, is the property to look for—and
589 coherence enters as the physical condition under which compatibility can be
590 maintained, which is what E78 asserts.

591 Second, uniqueness follows from the construction rather than from a sep-
592 arate selection principle. Accounts of conscious unity need something that
593 picks one experience from among the candidates a system could support;
594 integrated information theory postulates exclusion for this purpose (Tononi
595 et al., 2016). Here uniqueness arrives twice as a consequence: given a cover
596 whose local sections agree, the sheaf condition gives exactly one global sec-
597 tion, and given a contracting self-prediction map, Banach’s theorem gives
598 exactly one fixed point. The scope of that claim needs stating exactly, be-
599 cause the stronger reading is available and is not ours. Uniqueness is relative
600 to a cover and a map, and choosing them—which regions, which overlaps,
601 which encoding region, and so where the subject ends—is not supplied by ei-
602 ther theorem. It is what E78 and E89 carry. What the construction removes

603 is the need for an additional principle to select among the experiences a fixed
604 cover and map would otherwise admit; it does not remove the need to say
605 which cover and map a given subject has. That is a structural argument,
606 not evidence.

607 Third, the reflexive clause makes the self a condition rather than an addi-
608 tion. The predictive map restricts a global state to a region encoding it and
609 reads that encoding back out, so a fixed point is a state whose self-model is
610 accurate about the state it models. That condition is available only because
611 the state modelled is a single glued object: a system whose local descrip-
612 tions do not cohere has no such state for a self-model to be about. Unity
613 and selfhood are therefore one construction read at two levels, which is why
614 `UnifiedSelf` requires the fixed point to be the glued section itself rather
615 than an unrelated section.

616 The costs belong with the argument. The commitment does not address
617 why any physical condition is accompanied by experience; it is a proposal
618 about which physical state is the conscious one, not about why states of that
619 kind feel like anything. It is therefore open to the objection that a system
620 could satisfy every condition—compatible local states, a supercritical field,
621 an accurate self-model—while nothing is experienced. We have no reply to
622 that objection and claim none. The identification is also multiply realizable
623 as stated: nothing in the sheaf or fixed-point conditions is specific to cortex
624 or to electromagnetic coupling, so the substrate claim is carried entirely by
625 E56 and E67 and is empirical. No conclusion about artificial systems follows

626 from the formal results.

627 What could count against the commitment is dissociation. It predicts that
628 the two clauses and the presence of a unified conscious episode go together,
629 so a state in which local cortical descriptions are demonstrably compatible
630 and the field supercritical, while the subject reports no unified experience,
631 would tell against it; so would unified reported experience where compati-
632 bility demonstrably fails. Anesthetic emergence, dreaming and dissociative
633 states are the places to look, because they separate coherence from report
634 in ways ordinary waking does not. Such tests need an operational measure
635 of overlap compatibility on neural data, which this article does not supply;
636 constructing one is the main obstacle to making the commitment empirically
637 live rather than merely stated. Supplemental Material specifies what such an
638 observable would have to satisfy—a restriction map, a statistic with a null
639 under decoding noise, validation against configurations whose compatibility
640 is built rather than inferred—and the three ways it would fail silently, one
641 of which is reading compatibility off synchrony and so assuming away the
642 distinction being tested.

643 The physical tests of Sections 3 and 4 are independent of all of this. They
644 concern coupling, coherence and recovery, and can fail while the identification
645 stands, or succeed while it is wrong.

646 7 Discussion

647 The framework connects thermodynamic information constraints, collective
648 phase organization and compatible local representations, conditionally and
649 with its assumptions named. Its biological content is a coupling-gated re-
650 covery hypothesis with defined observables, a stated protocol, and failure
651 conditions declared in advance. Its formal content is a composition whose
652 eight gaps are exhibited rather than argued past, and which can be inhabited
653 by a single model. Its interpretive content is the identification of Section 6,
654 offered as a commitment with reasons.

655 The simulations are useful because they restrict the hypothesis rather
656 than support it. Finite ramps smear the stationary onset to the point where
657 a fitted onset exponent chiefly reports the ramp; an unpinned collective
658 phase makes independence diagnostics ambiguous in exactly the supercritical
659 regime the model cares about; and squared-drift plasticity maintains coher-
660 ence while degrading structural alignment. The second and third of these
661 are results about method and about a common intuition, and they transfer
662 beyond this framework.

663 The proposal differs in emphasis from accounts based on integrated in-
664 formation (Tononi et al., 2016), global workspace dynamics (Baars, 1997;
665 Dehaene et al., 2014), or predictive processing (Friston, 2010). It specifies a
666 candidate interaction medium and a mathematical compatibility condition,
667 and, relative to a given cover and self-prediction map, it derives the unique-

ness those accounts postulate. It does not derive the exclusion principle of integrated information, identify every broadcast with a field transition, or derive a generative model from a dissipation bound. Continuous fields can carry incompatible local descriptions and discrete models can admit sheaf constructions, so continuity alone does not settle the combination problem; compatibility is what this account offers in its place.

Two formal results bear on the substrate question, and both admit a stronger reading than they support. At matched coupling resource, a fixed wiring support that excludes a better-correlated pair can be suboptimal—a statement about constrained optimization, not about biology. On an atomless substrate, a kernel supported on finitely many points contributes zero to the chosen continuum coupling integral. That is a property of the measure and the observable, and it cuts against this framework’s own coarse-graining step as much as against anything else. It does not imply that digital hardware cannot approximate a field or support consciousness, and no biological-versus-artificial conclusion follows.

The empirical priorities follow directly from Section 3.2: estimate effective coupling and diffusion in consistent units, test the proposed relation between coupling and extracellular geometry, and measure recovery against an independently defined behavioral endpoint under matched model comparison. A favorable concentration–coherence relation would support a statistical description, not identify its physical cause; an unfavorable result in a regime satisfying the stated assumptions would constrain that description.

691 Successful synchronization simulations cannot substitute for measurements
692 distinguishing ephaptic interactions from other coupling mechanisms.

693 The article therefore asks to be judged in three separable parts: a pre-
694 diction that can be tested, a composition that can be checked, and an iden-
695 tification that can be argued about. Keeping them separate is what makes
696 the whole assessable.

697 **Data and code availability**

698 The Lean development, simulation scripts, tests and saved numerical sum-
699 maries are maintained in the manuscript source repository. Publication sim-
700 ulation values are regenerated from saved summaries without rerunning pro-
701 duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset
702 ds005620 (Bajwa et al., 2025); preprocessing and selection are described
703 in Supplemental Material. The source repository contains the reproducible
704 analysis and does not redistribute the raw recordings.

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706 This research received no external funding. The author declares no compet-
707 ing interests.

708 Use of AI tools

709 AI assistance to this work was substantial and is not separable by task.
710 Several coding assistants and chat interfaces were used in an interleaved way
711 throughout: Google’s Antigravity CLI (Gemini 3.1, Claude 4.6 Opus, Claude
712 4.6 Sonnet), OpenAI’s Codex CLI (Astra, Terra, Luna, Sol), Claude Code
713 (Claude Opus 5), Gemini Pro through its web interface, and Fable (Claude
714 Opus 4) through OpenRouter, accessed with the Hermes CLI. Between them
715 they contributed to the Lean development, the simulation and analysis code,
716 and the drafting and revision of both documents. Because the tools were
717 used in combination and in alternation, no section, proof, figure or result is
718 attributable to any one of them, and none is claimed to be. The author is
719 responsible for all content, including everything a tool produced. Machine
720 checking establishes the stated Lean results under their hypotheses; it does
721 not validate biological interpretations or replace scientific review of generated
722 code and prose.

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